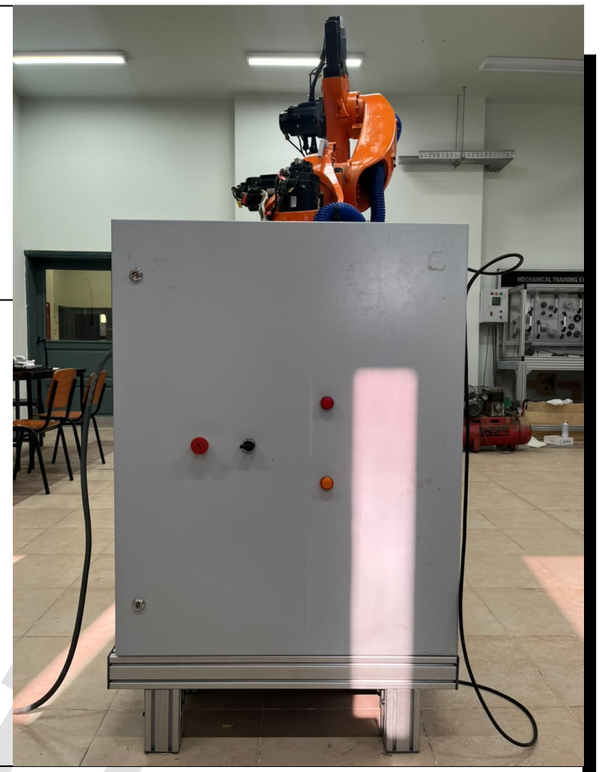
Assiut University

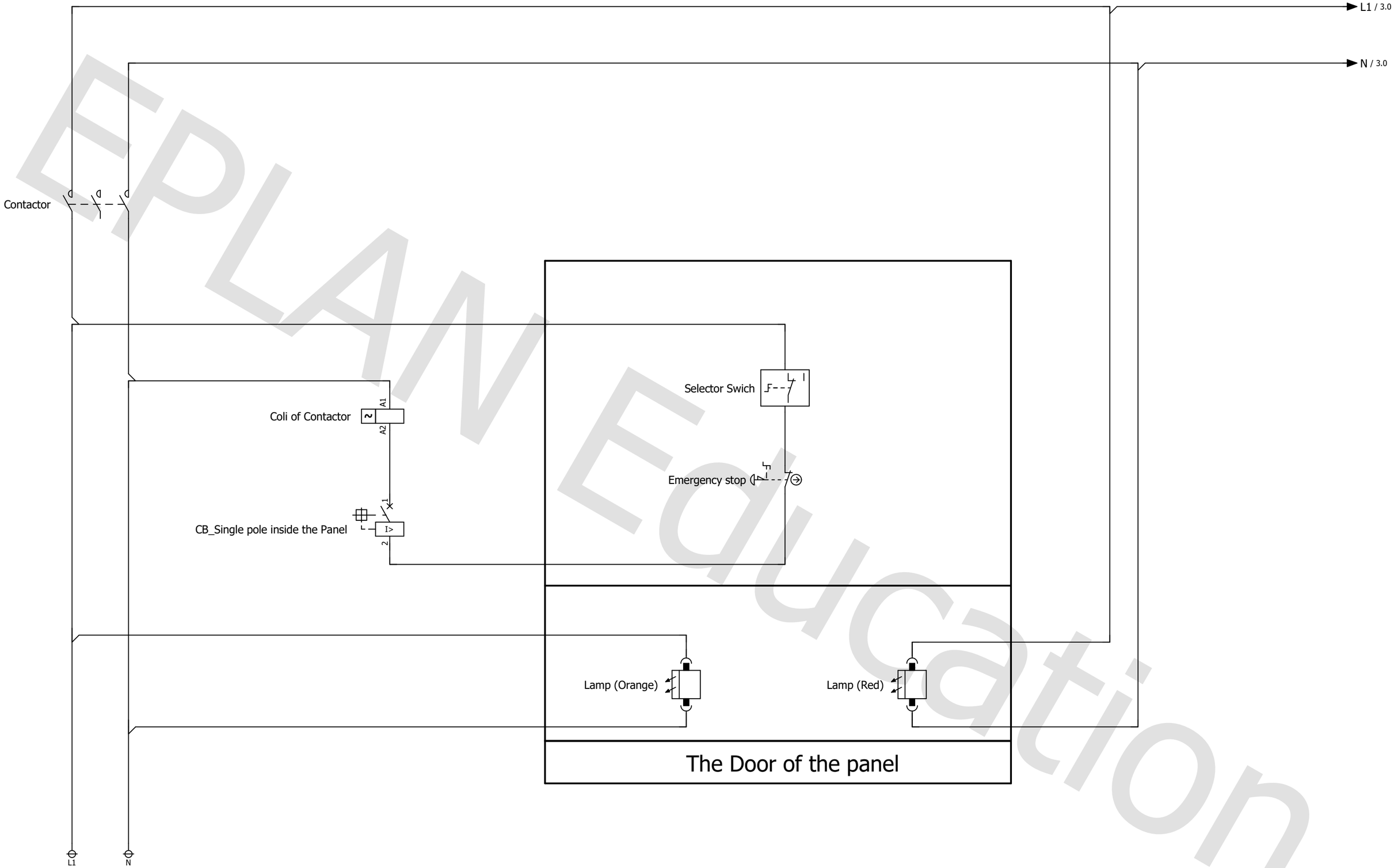
Faculty of Engineering
Assiut
Phone

Company / customer	
Project description	Motion Control solution for Industrial Manipulator
Job number	KP-2026-001
Commission	Graduation Project
Manufacturer (company)	Assiut University
Path	KUKA Robot Control Panel
Project name	Kuka Robot Arm Panel
Make	Schematic diagram
Type	Electric Contol Planel
Place of installation	Robotics Lab _ AITU
Responsible for project	Kuka Robot Team
Part feature	



Created on	7/31/2026		
Edit date	8/4/2026	by (short name) Abdullah Rabea	Number of pages 10

EPLAN Education



EPLAN Education

			Date	8/3/2026	Graduation Project	Assiut University	Main Power and Safety circuit diagram		= CA1
			Ed.	Abdullah Rabea					+ EAA
			Appr.		Motion Control solution for Industrial Manipulator			KP-2026-001	Page 2
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L1 / 2.9

N / 2.9

L1 / 4.0

N / 4.0

CB1_A1 (M)
Circuit breaker of joint 1
I> I> I>

9.7 L Brake 1

9.7 N Brake 1

L1
L2
Main Power

L1C
L2C
Main Power

U
V
W
Motor Power

PE

Digital operator

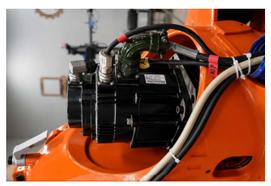
Host Controller Cable 50 pins cable connector

A1 _50 Pins / 10.9

Encoder connector
1 2 3 4 5 6 7 8 9 10

Motor drive control A1

A B C PE
M
3 ~



Encoder connector A1
1 2 3 4 5 6 7 8 9 10

Brake Cable | Motor Cable | Encoder cable

Joint of A1 3 Cables

EPLAN Education

2 4

			Date	8/3/2026	Graduation Project	Assiut University	A1_Motor 1		= CA1	
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L1 / 3.9

N / 3.9

L1 / 5.0

N / 5.0

CB2_A2 (M)
Circuit breaker of joint 2
I> I> I>

9.7 L Brake 2

9.7 N Brake 2

L1
L2
Main Power

L1C
L2C
Main Power

U
V
W
Motor Power

PE

Digital operator

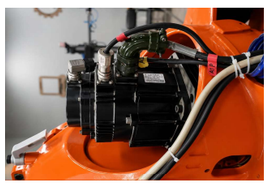
Host Controller Cable 50 pins cable connector

A2_ 50Pins / 10.8

Encoder connector
1 2 3 4 5 6 7 8 9 10

Motor drive control A2

A B C PE
M
3 ~



Encoder connector A2
1 2 3 4 5 6 7 8 9 10

Brake Cable | Motor Cable | Encoder cable

Joint of A2 3 Cables

EPLAN Education

3 5

			Date	8/4/2026	Graduation Project	Assiut University	A2_Motor 2	= CA1	
			Ed.	Abdullah Rabea				+ EAA	
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L1 / 4.9

N / 4.9

L1 / 6.0

N / 6.0

CB3_A3 (M)
Circuit breaker of joint 3

I> I> I>

9.8
L Brake 3

9.8
N Brake 3

L1
L2
Main Power

L1C
L2C
Main Power

U
V
W
Motor Power

PE

Digital operator

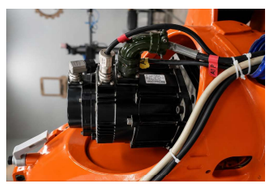
Host Controller Cable 50 pins cable connector

A3 _50 Pins / 10.7

Encoder connector
1 2 3 4 5 6 7 8 9 10

Motor drive control A3

A B C PE
M
3 ~



Encoder connector A1
1 2 3 4 5 6 7 8 9 10

Brake Cable | Motor Cable | Encoder cable

Joint of A3 3 Cables

EPLAN Education

4 6

			Date	8/3/2026	Graduation Project		Assiut University	A3_Motor 3	= CA1	
			Ed.	Abdullah Rabea					+ EAA	
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L1 / 5.9

N / 5.9

L1 / 7.0

N / 7.0

CB4_A4 (M)
Circuit breaker of joint 4

I> I> I>

9.8
L Brake 4

9.8
N Brake 4

L1
L2
L3
Main Power

36-pin cable connector

A4 _36 Pins / 10.7

U
V
W
PE
Motor Power

Encoder connector
1 2 3 4 5 6 7 8 9 10

Motor drive control A4

A B C PE
M
3 ~



Encoder connector A1
1 2 3 4 5 6 7 8 9 10

Brake Cable | Motor Cable | Encoder cable

Joint of A4 3 Cables

EPLAN Education

5 7

			Date	8/3/2026	Graduation Project	Assiut University	A4_Motor 4		= CA1	
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L1 / 6.9

N / 6.9

L1 / 8.0

N / 8.0

CB5_A5
Circuit breaker of joint 5

I> I> I>

9.8
L Brake 5

9.9
N Brake 5

L1
L2
L3
Main Power

U
V
W
PE
Motor Power

Motor drive control A5

36-pin cable connector

A5_36 Pins / 10.6

Encoder connector

1 2 3 4 5 6 7 8 9 10

A B C PE

M
3 ~

Brake Cable

Motor Cable

Encoder cable

Encoder connector A1

1 2 3 4 5 6 7 8 9 10

Joint of A5 3 Cables

EPLAN Education

6 8

			Date	8/3/2026	Graduation Project		Assiut University	A5_Motor 5		= CA1
			Ed.	Abdullah Rabea						+ EAA
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L1 / 7.9

N / 7.9

L1 /

N /

CB6_A6 (M)
Circuit breaker of joint 6

I> I> I>

9.9
L Brake 6

9.9
N Brake 6

L1
L2
L3
Main Power

36-pin cable connector

A6 _36 Pins / 10.5

U
V
W
PE
Motor Power

Encoder connector

1 2 3 4 5 6 7 8 9 10

Motor drive control A6

A B C PE

M
3 ~



Encoder connector A1

1 2 3 4 5 6 7 8 9 10

Brake Cable | Motor Cable | Encoder cable

Joint of A6 3 Cables

EPLAN Education

7 9

			Date	8/3/2026	Graduation Project		Assiut University	A6_Motor 6		= CA1
			Ed.	Abdullah Rabea						+ EAA
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EDUCATION Product

L1 / 9.9

N / 9.9

L1 / 9.0

N / 9.0

Power supply

24 volt

L

N

V+

V-

3.0 N Brake 1
4.0 N Brake 2
5.0 N Brake 3
6.0 N Brake 4
7.0 N Brake 5
8.0 N Brake 6

3.0 L Brake 1
4.0 L Brake 2
5.0 L Brake 3
6.0 L Brake 4
7.0 L Brake 5
8.0 L Brake 6

5 Volt Input

24 Volt input

Brake A1
Brake A2
Brake A3
Brake A4
Brake A5
Brake A6

the connector of the signal brake
This schematic shows

B6 5V 10.2
B5 5V 10.1
B4 5V 10.1
B3 5V 10.1
B2 5V 10.1
B1 5V 10.0
G 10.0

Power supply

5 Volt

L

N

V+

V-

8

10

			Date	8/3/2026	Graduation Project	Assiut University	Brakes circuit	= CA1	
			Ed.	Abdullah Rabea				+ EAA	
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L1 / 10.9

N / 10.9

L1 / 10.0

N / 10.0

Power supply
12 volt

L
N
V+
V-

PCB Power Supply

12 Volt
Input

CANBus
Connector

9.6 G
9.6 B1 5V
9.6 B2 5V
9.6 B3 5V
9.5 B4 5V
9.5 B5 5V
9.5 B6 5V

This schematic shows
the connector of the signal brake

CANBus
Connector

A6 8.7 36 Pins
A5 7.7 36 Pins
A4 6.7 36 Pins
A3 5.7 50 Pins
A2 4.7 50 Pins
A1 3.7 50 Pins

EPLAN Education

			Date	8/3/2026	Graduation Project	Assiut University	PCB Box	= CA1	
			Ed.	Abdullah Rabea				+ EAA	
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